ChE 311

HW # 2

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1. a) $V = [by, 0, 0]$ $\nabla = [\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}]$

$\nabla V = \sum_i \sum_j \delta_i \delta_j \nabla_i V_j$

$= \delta_1\delta_1 \nabla_1 V_1 + \delta_1\delta_2 \nabla_1 V_2 + \delta_1\delta_3 \nabla_1 V_3 + \ldots + \delta_3\delta_3\nabla_3 V_3$

$= 0\,\delta_1\delta_1 + 0\,\delta_1\delta_2 + 0\,\delta_1\delta_3 + b\,\delta_2\delta_1 + 0\,\delta_2\delta_2 + 0\,\delta_2\delta_3 + 0\,\delta_3\delta_1 + 0\,\delta_3\delta_2 + 0\,\delta_3\delta_3$

$\nabla V = \begin{Bmatrix} 0 & 0 & 0 \\ b & 0 & 0 \\ 0 & 0 & 0 \end{Bmatrix}$

$\rho VV = \rho(VV) = \rho \sum_i \sum_j \delta_i \delta_j V_i V_j = \rho \begin{Bmatrix} b^2y^2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{Bmatrix} = \begin{Bmatrix} \rho b^2 y^2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{Bmatrix}$

b) $V = [by, bx, 0]$ $\nabla = [\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}]$

$\nabla V = \sum_i \sum_j \delta_i \delta_j \nabla_i V_j$

$= 0\,\delta_1\delta_1 + b\,\delta_1\delta_2 + 0\,\delta_1\delta_3 + b\,\delta_2\delta_1 + 0\,\delta_2\delta_2 + 0\,\delta_2\delta_3 + 0\,\delta_3\delta_1 + 0\,\delta_3\delta_2 + 0\,\delta_3\delta_3$

$= \begin{Bmatrix} 0 & b & 0 \\ b & 0 & 0 \\ 0 & 0 & 0 \end{Bmatrix}$

$\rho VV = \rho(VV) = \rho(VV) = \rho \sum_i \sum_j \delta_i \delta_j V_i V_j = \rho \begin{Bmatrix} b^2y^2 & by\,bx & 0 \\ bx\,by & b^2x^2 & 0 \\ 0 & 0 & 0 \end{Bmatrix} = \begin{Bmatrix} \rho b^2y^2 & \rho\,by\,bx & 0 \\ \rho\,bx\,by & \rho b^2x^2 & 0 \\ 0 & 0 & 0 \end{Bmatrix}$

c) $V = [-by, bx, 0]$ $\nabla = [\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}]$

$\nabla V = \sum_i \sum_j \delta_i \delta_j \nabla_i V_j$

$= 0\,\delta_1\delta_1 + b\,\delta_1\delta_2 + 0\,\delta_1\delta_3 - b\,\delta_2\delta_1 + 0\,\delta_2\delta_2 + 0\,\delta_2\delta_3 + 0\,\delta_3\delta_1 + 0\,\delta_3\delta_2 + 0\,\delta_3\delta_3$

$= \begin{Bmatrix} 0 & b & 0 \\ -b & 0 & 0 \\ 0 & 0 & 0 \end{Bmatrix}$

$\rho VV = \rho(VV) = \rho \sum_i \sum_j \delta_i \delta_j V_i V_j = \rho \begin{Bmatrix} b^2y^2 & -by\,bx & 0 \\ -bx\,by & b^2x^2 & 0 \\ 0 & 0 & 0 \end{Bmatrix} = \begin{Bmatrix} \rho b^2y^2 & -\rho\,by\,bx & 0 \\ -\rho\,bx\,by & \rho b^2x^2 & 0 \\ 0 & 0 & 0 \end{Bmatrix}$

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d) $V = [-0.5bx, -0.5by, bz]$ $\nabla = [\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}]$

$\nabla V = \sum_i \sum_j \delta_i \delta_j \nabla_i V_j$

$= -0.5b\,\delta_1\delta_1 + 0\,\delta_1\delta_2 + 0\,\delta_1\delta_3 + 0\,\delta_2\delta_1 - 0.5b\,\delta_2\delta_2 + 0\,\delta_2\delta_3$
$+ 0\,\delta_3\delta_1 + 0\,\delta_3\delta_2 + b\,\delta_3\delta_3$

$$= \begin{Bmatrix} -0.5b & 0 & 0 \\ 0 & -0.5b & 0 \\ 0 & 0 & b \end{Bmatrix}$$

$$\rho vv = \rho(vv) = \rho \sum_i \sum_j \delta_i \delta_j v_i v_j = \rho \begin{Bmatrix} \frac{1}{4}b^2x^2 & \frac{1}{4}bxby & -\frac{1}{2}bxbz \\ \frac{1}{4}bxby & \frac{1}{4}b^2y^2 & -\frac{1}{2}bybz \\ -\frac{1}{2}bxbz & -\frac{1}{2}bybz & b^2z^2 \end{Bmatrix}$$

$$= \begin{Bmatrix} \frac{1}{4}\rho b^2x^2 & \frac{1}{4}\rho bxby & -\frac{1}{2}\rho bxbz \\ \frac{1}{4}\rho bxby & \frac{1}{4}\rho b^2y^2 & -\frac{1}{2}\rho bybz \\ -\frac{1}{2}\rho bxbz & -\frac{1}{2}\rho bybz & \rho b^2z^2 \end{Bmatrix}$$

2. $\mu = 30\ Pa \cdot s$, $W_{out} = 5°/s$ $L = 1m$ $R_{out} = 1m$ $d = 0.005m$ $R_{in} = 1 - 0.005 = 0.995m$

$\omega = \frac{V}{R} = \frac{5°/s}{180°} \pi\ rad = 0.0873\ rad/s$

$V = \omega \cdot R = 0.0873\ rad/s \cdot 1m = 0.0873\ m/s$

$\frac{F}{A} = \mu \frac{\Delta V}{Y} = 30\ Pa \cdot s \cdot \frac{0.0873\ m/s}{1m}$

$A = 2\pi r L = 2\pi (1m)(1m) = 6.283\ m^2$

$F = A \cdot \mu \frac{\Delta V}{Y} = 6.283m^2 \cdot 30\ Pa \cdot s \frac{0.0873\ m/s}{0.005m} = 3291\ N$

$T = F \cdot R \sin\theta = 3291\ N \times 1m \times \sin 90° = 3291\ N \cdot m = 3291\ J$